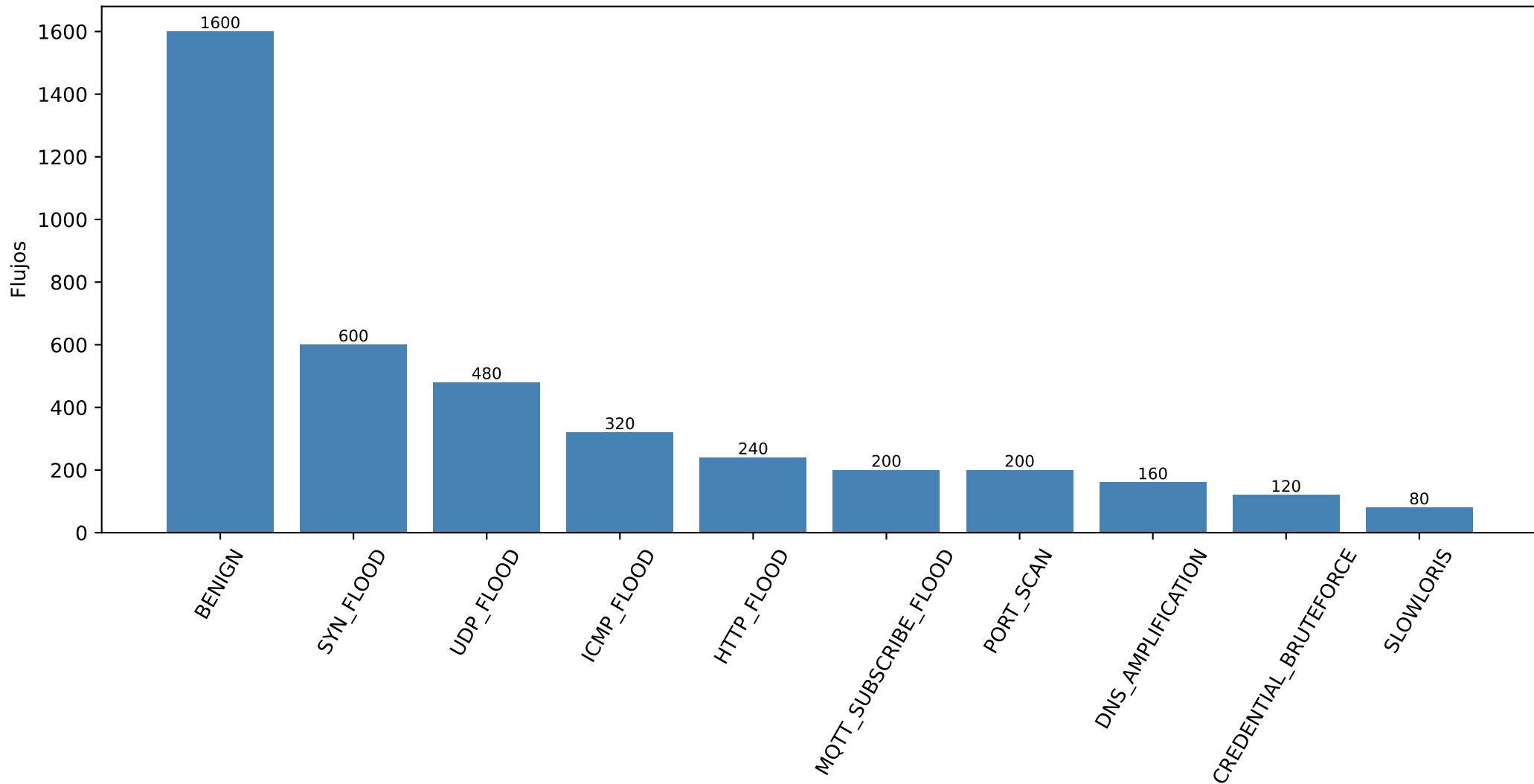


IoT-SDN-AI — Reporte de evaluación

run: run_synth

Dataset:	5,000 flujos
Features:	46
Clases:	10 (BENIGN, CREDENTIAL_BRUTEFORCE, DNS_AMPLIFICATION, HTTP_FLOOD, ICMP_FLOOD...)
Modelos:	RandomForest, XGBoost, MLP, IsolationForest, AutoEncoder
Estrategia:	stratified train/test split 80/20, scaler StandardScaler

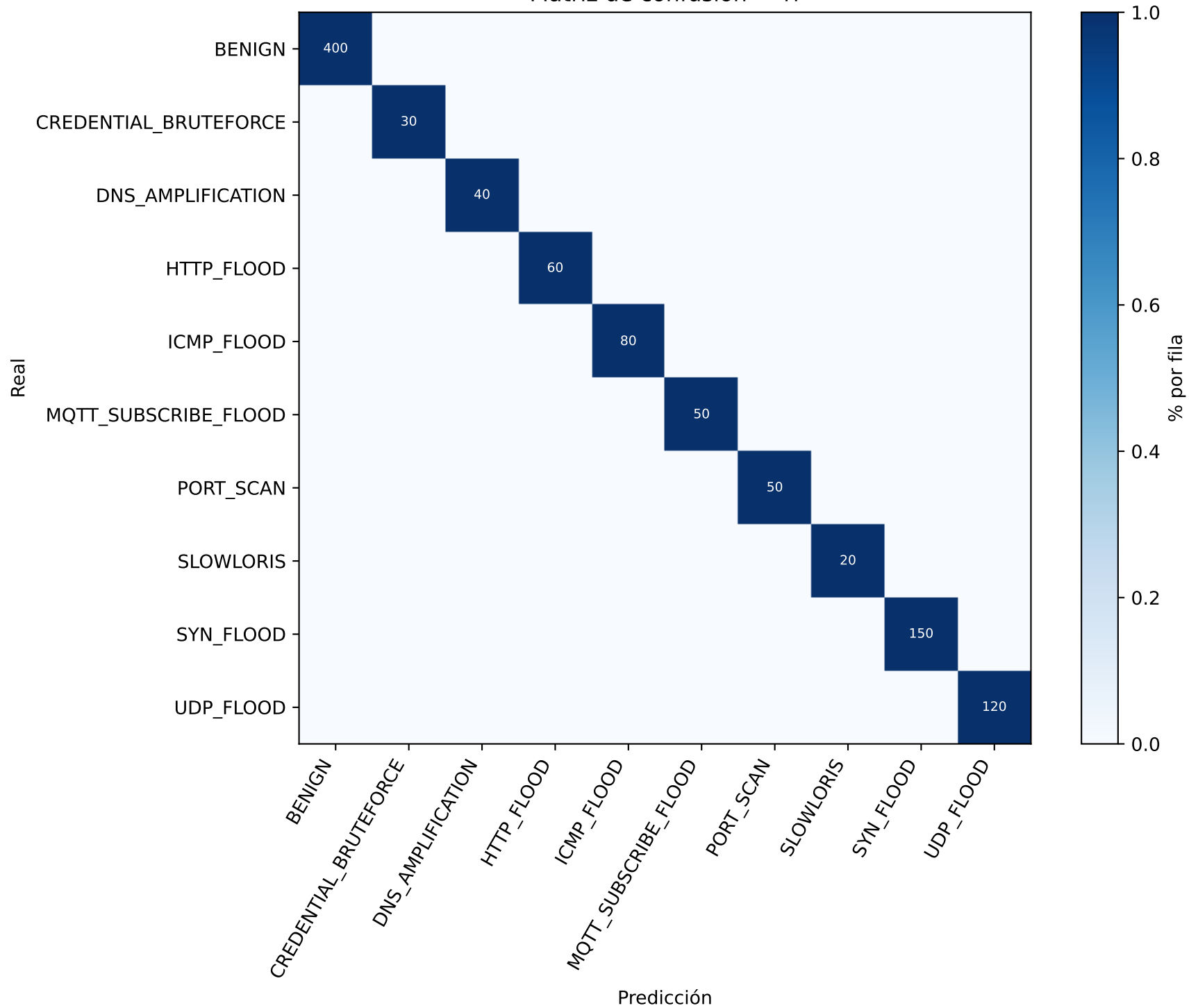
Distribución de clases en el dataset



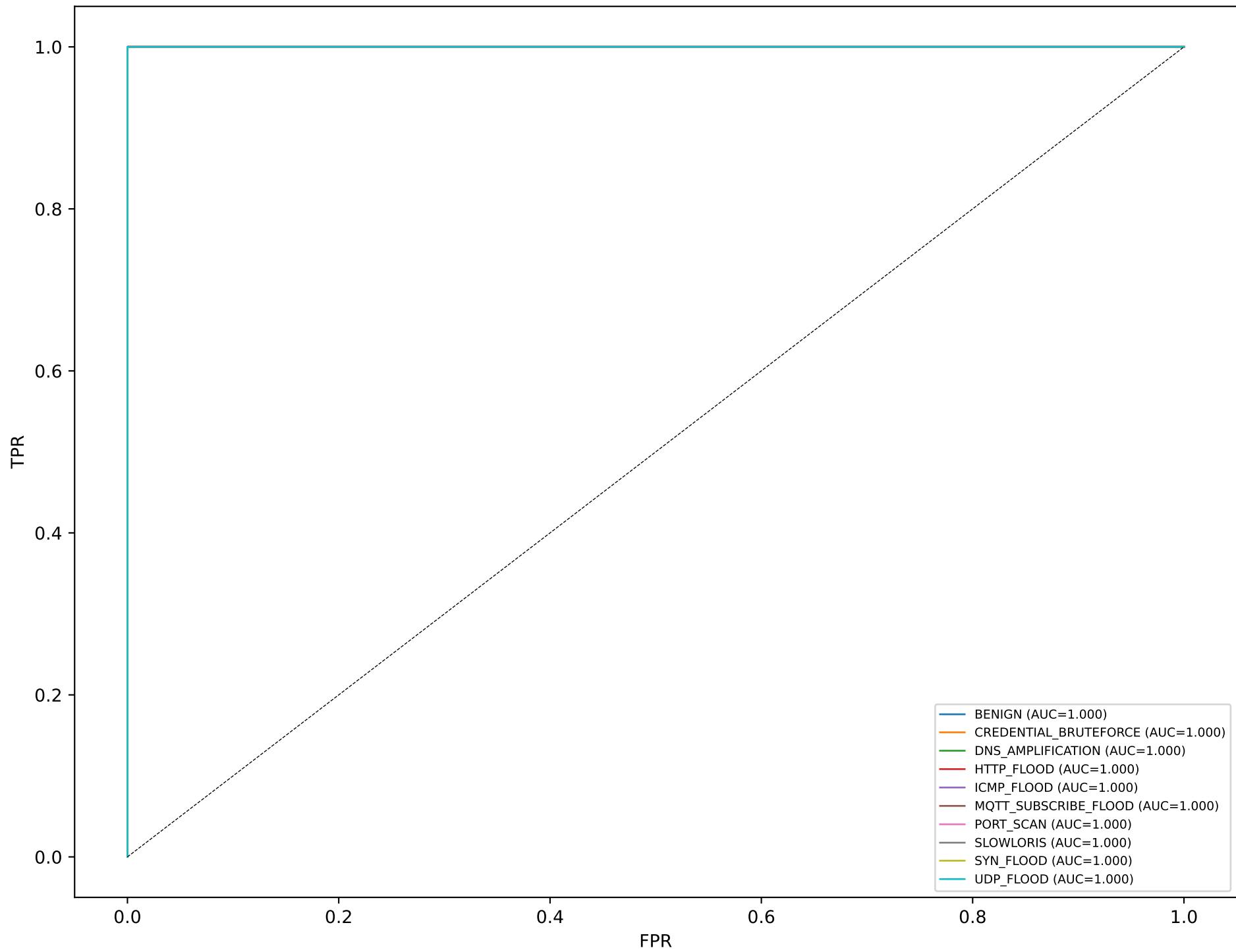
Classification report — rf

	precision	recall	f1-score	support
BENIGN	1.00	1.00	1.00	400
CREDENTIAL_BRUTEFORCE	1.00	1.00	1.00	30
DNS_AMPLIFICATION	1.00	1.00	1.00	40
HTTP_FLOOD	1.00	1.00	1.00	60
ICMP_FLOOD	1.00	1.00	1.00	80
MQTT_SUBSCRIBE_FLOOD	1.00	1.00	1.00	50
PORT_SCAN	1.00	1.00	1.00	50
SLOWLORIS	1.00	1.00	1.00	20
SYN_FLOOD	1.00	1.00	1.00	150
UDP_FLOOD	1.00	1.00	1.00	120
accuracy			1.00	1000
macro avg	1.00	1.00	1.00	1000
weighted avg	1.00	1.00	1.00	1000

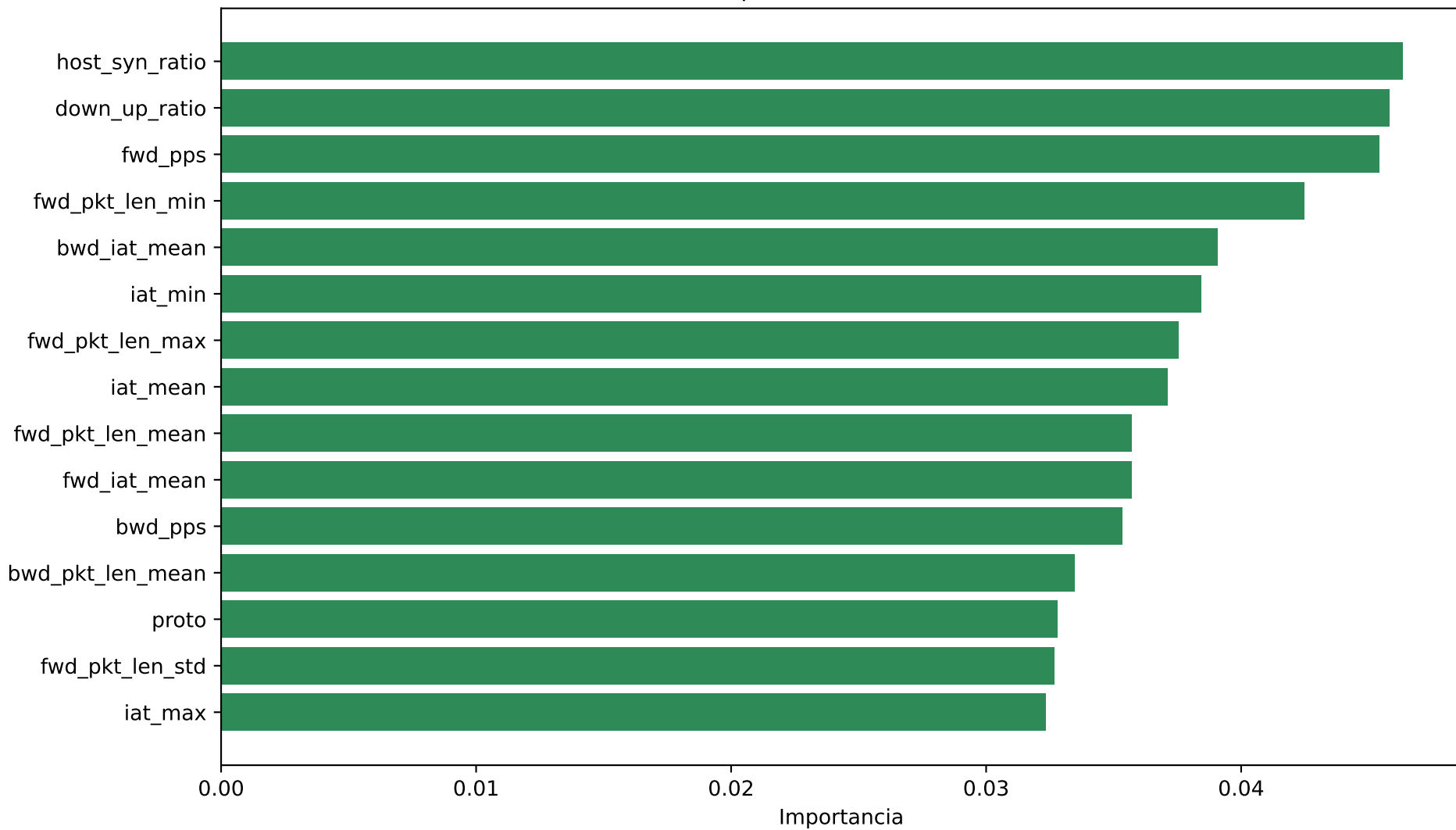
Matriz de confusión — rf



ROC OvR — rf



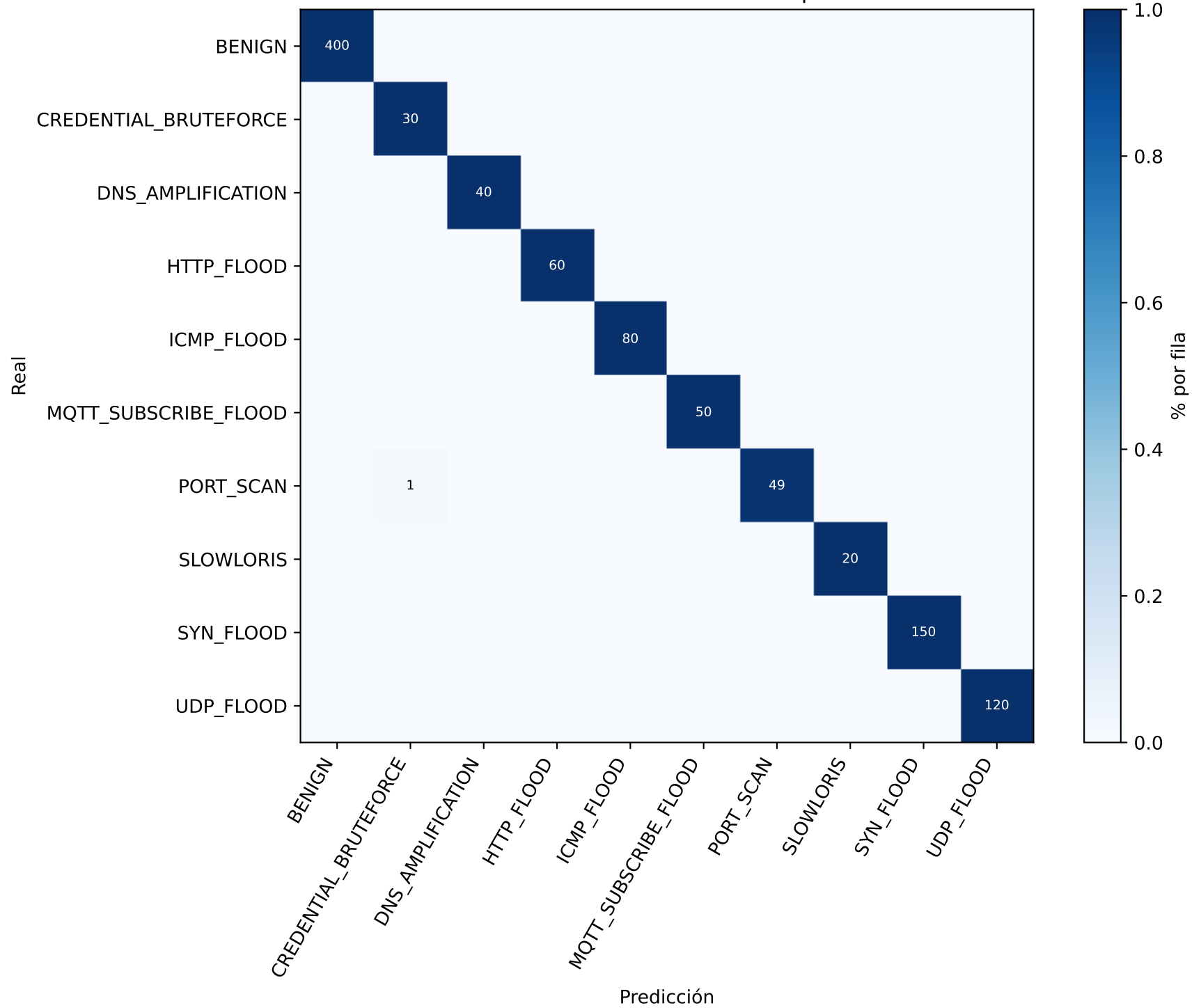
Top-15 features — rf



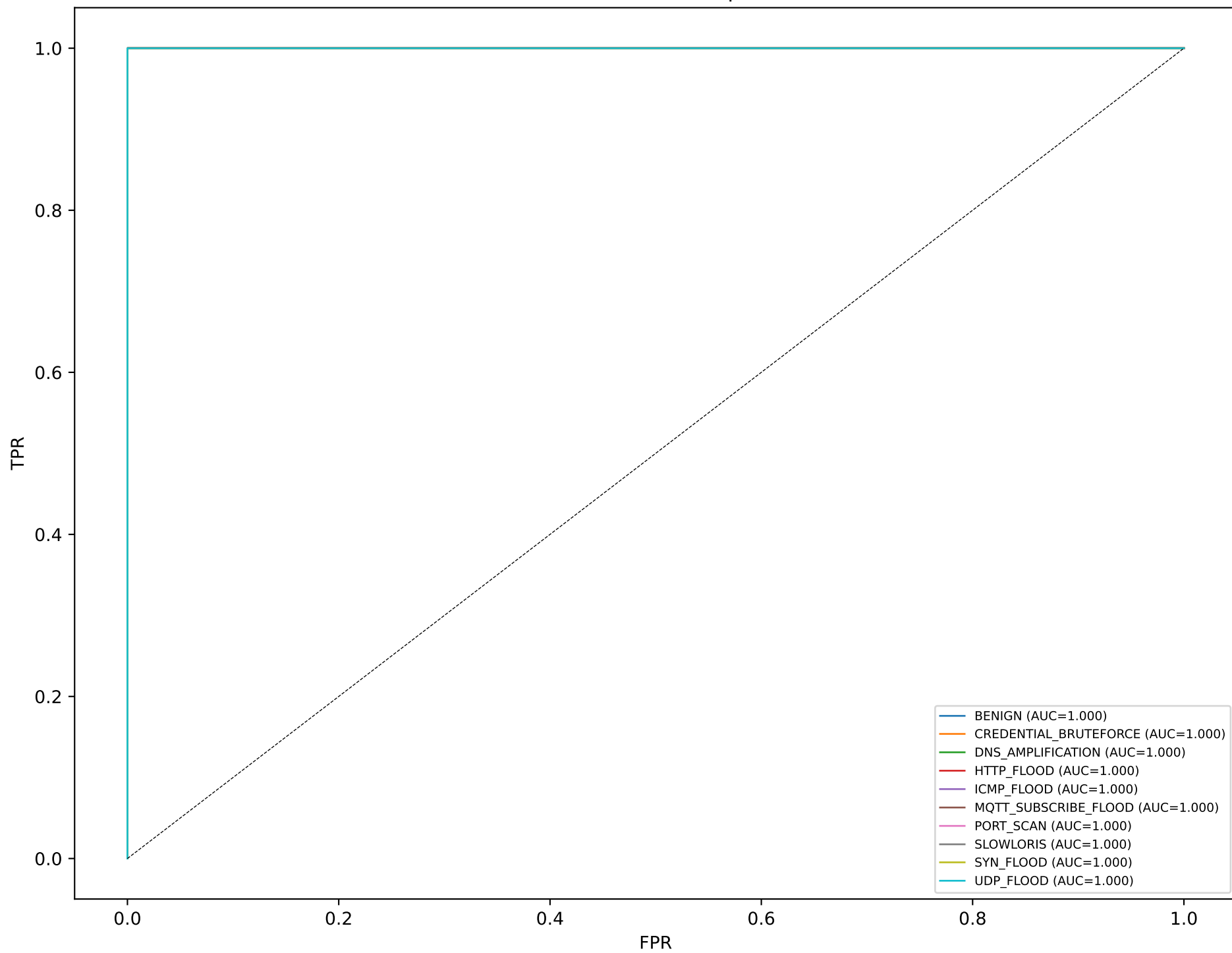
Classification report — mlp

	precision	recall	f1-score	support
BENIGN	1.00	1.00	1.00	400
CREDENTIAL_BRUTEFORCE	0.97	1.00	0.98	30
DNS_AMPLIFICATION	1.00	1.00	1.00	40
HTTP_FLOOD	1.00	1.00	1.00	60
ICMP_FLOOD	1.00	1.00	1.00	80
MQTT_SUBSCRIBE_FLOOD	1.00	1.00	1.00	50
PORT_SCAN	1.00	0.98	0.99	50
SLOWLORIS	1.00	1.00	1.00	20
SYN_FLOOD	1.00	1.00	1.00	150
UDP_FLOOD	1.00	1.00	1.00	120
accuracy			1.00	1000
macro avg	1.00	1.00	1.00	1000
weighted avg	1.00	1.00	1.00	1000

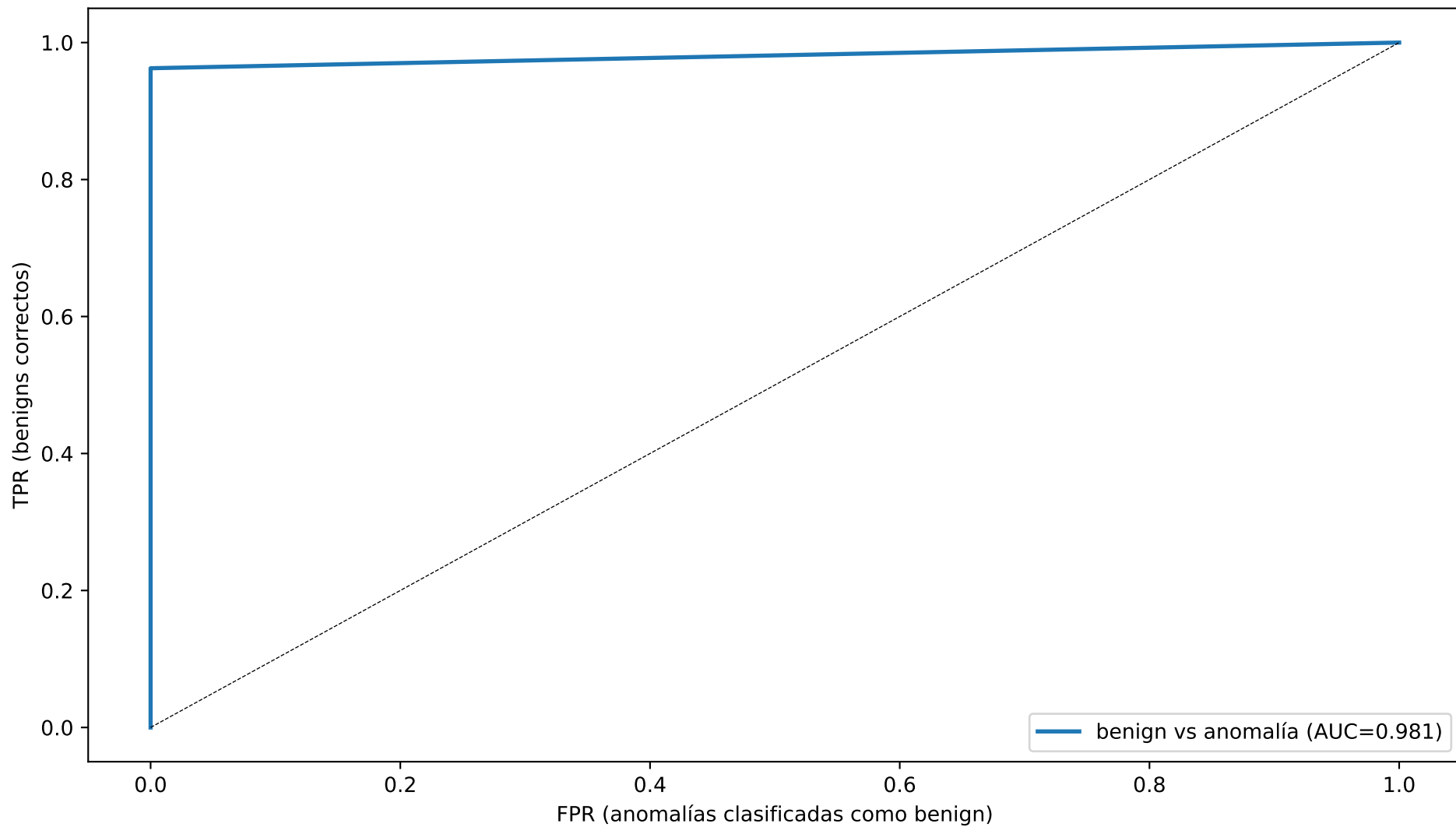
Matriz de confusión — mlp



ROC OvR — mlp



ROC binaria — isoforest



ROC binaria — autoencoder

